

Lab03 TCP

2. A first look at the captured trace

1. What is the IP address and TCP port number used by the client computer (source) that is transferring the file to gaia.cs.umass.edu? To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the "details of the selected packet header window" (refer to Figure 2 in the "Getting Started with Wireshark" Lab if you're uncertain about the Wireshark windows).

IP address is 192.168.68.65 and the source TCP port is 56739

No.	Time	Source	Destination	Protocol	Length	Info
270	00:05:18.711929	192.168.68.65	128.119.245.12	HTTP	1092	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1
309	00:05:18.890268	128.119.245.12	192.168.68.65	HTTP	843	HTTP/1.1 200 OK (text/html)

> Frame 270: 1092 bytes on wire (8736 bits), 1092 bytes captured (8736 bits) on interface en0, id 0	Ethernet
> Ethernet II, Src: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)	0 15 16
> Internet Protocol Version 4, Src: 192.168.68.65, Dst: 128.119.245.12	Destination TPLink_9f:6a:ac
> Transmission Control Protocol, Src Port: 56739, Dst Port: 80, Seq: 148410, Ack: 1, Len: 1026	Source 12:56:2d:e7:53:95
Source Port: 56739	Type
Destination Port: 80	
[Stream index: 21]	
> [Conversation completeness: Incomplete (12)]	
[TCP Segment Len: 1026]	
Sequence Number: 148410 (relative sequence number)	
Sequence Number (raw): 2925405562	
[Next Sequence Number: 149436 (relative sequence number)]	

2. What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?

The IP address of gaia.cs.umass.edu is: 128.119.245.12, port number 80 is used for sending and receiving TCP segments.

3. What is the IP address and TCP port number used by your client computer (source) to transfer the file to gaia.cs.umass.edu?

Client (Source) IP Address: 192.168.68.65

Client (Source) TCP Port Number: 56739

3. TCP Basics

4. What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? What is it in the segment that identifies it as an SYN segment?

The sequence number of the TCP SYN segment that is used to initiate the TCP connection is

0(relative sequence number) or a raw sequence number 18317345; We can see from below that the segment that identified the segment as a SYN segment is the segment that identified the segment as a SYN segment.

179

13:44:53.473676

192.168.68.65

20.42.73.26

TLSv1.3

985

Application Data

Seq=6507 Ack=1494 Win=4194816 Len=0 TSval=55199963 TSecr=2017525

180

13:44:53.578759

20.42.73.26

192.168.68.65

TCP

66

443 -> 61112 [ACK]

Seq=6507 Ack=6672 Win=138880 Len=0 TSval=2017525522 TSecr=551999

181

13:44:53.578761

20.42.73.26

192.168.68.65

TLSv1.3

169

Application Data

Seq=6507 Ack=3841 Win=4194816 Len=0 TSval=55199963 TSecr=2017525

182

13:44:53.578763

20.42.73.26

192.168.68.65

TLSv1.3

128

Application Data

Seq=6507 Ack=3841 Win=4194816 Len=0 TSval=55199963 TSecr=2017525

183

13:44:53.578766

20.42.73.26

192.168.68.65

TCP

66

443 -> 61112 [ACK]

Seq=6507 Ack=3841 Win=4194816 Len=0 TSval=55199963 TSecr=2017525

184

13:44:53.571820

192.168.68.65

20.42.73.26

TCP

66

61112 -> 443 [ACK]

Seq=3841 Ack=6672 Win=138880 Len=0 TSval=2017525522 TSecr=551999

185

13:44:53.571647

192.168.68.65

20.42.73.26

TLSv1.3

97

Application Data

Seq=3841 Ack=6672 Win=138880 Len=0 TSval=2017525522 TSecr=551999

187

13:44:53.665989

20.42.73.26

192.168.68.65

TLSv1.3

375

Application Data, Application Data

Seq=3872 Ack=6981 Win=130752 Len=0 TSval=2017525617 TSecr=552008

188

13:44:53.666225

20.42.73.26

192.168.68.65

TCP

66

61112 -> 443 [ACK]

Seq=3872 Ack=6981 Win=130752 Len=0 TSval=2017525617 TSecr=552008

189

13:44:53.719761

20.42.73.26

192.168.68.65

TCP

66

443 -> 61112 [ACK]

Seq=6981 Ack=3872 Win=4194816 Len=0 TSval=55200114 TSecr=2017525

191

13:44:53.775932

192.168.68.62

192.168.68.65

TCP

78

51967 -> 7000 [SYN]

Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=2795619953 TSecr=0

193

13:44:53.773748

192.168.68.65

192.168.68.62

TCP

78

7000 -> 51967 [SYN, ACK]

Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=64 TSval=183071065

Ethernet II, Src: 8a:8e:01:97:c8:71 (8a:8e:01:97:c8:71), Dst: 12:56:2d:e7:53:95 (12:56:2d:e7:53:95)

Internet Protocol Version 4, Src: 192.168.68.62, Dst: 192.168.68.65

Transmission Control Protocol, Src Port: 51967, Dst Port: 7000, Seq: 0, Len: 0

Source Port: 51967

Destination Port: 7000

[Stream index: 10]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 18317345

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 0

Acknowledgment number (raw): 0

1011.... = Header Length: 44 bytes (11)

Flags: 0x002 (SYN)

0000.... = Reserved: Not set

...0.... = Accurate ECN: Not set

...0.... = Congestion Window Reduced: Not set

....0.... = ECU-Echo: Not set

....0.... = Urgent: Not set

....0.... = Acknowledgment: Not set

....0.... = Push: Not set

....0.... = Reset: Not set

....0.... = Syn: Set

....0.... = Fin: Not set

[TCP Flags:S.]

Window: 65535

[Calculated window size: 65535]

Checksum: 0x5888 [unverified]

[Checksum Status: Unverified]

Ethernet

Destination 12:56:2d:e7:53:95

Source 8a:8e:01:97:c8:71

Type IPv4

Internet Protocol Version 4

Version 4	Header 20	Differentiated Service 0x00	Total Length 64
Identification 0x0000 (0)		Flags 0x2	Fragment Offset 0
Time to Live 64	Protocol TCP	Header Checksum 0x30eb	
Source Address 192.168.68.62			
Destination Address 192.168.68.65			

5. What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is the value of the Acknowledgment field in the SYNACK segment? How did gaia.cs.umass.edu determine that value? What is it in the segment that identifies the segment as a SYNACK segment?

The sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN is: Q

The value of the Acknowledgement field in the SYNACK segment is: 1.

The ACKnowledgement field value in the SYN-ACK segment from “gaia.cs.umass.edu” is determined by incrementing the initial sequence number of the SYN segment from the client computer by 1. The client's SYN segment starts with a sequence number of 0.

The SYN flag and Acknowledgement flag in the segment are set to 1 and they indicate that this segment is a SYNACK segment.

189 13:44:53.719701 20.42.73.20 192.168.68.65 TCP 66 443 → 8111 [ACK] Seq=6981 Ack=5872 Win=4194810 Len=0 TSval=35200114 TSecr=...

191 13:44:53.775832 192.168.68.62 192.168.68.65 TCP 78 51967 → 7000 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=2795619953 TSecr=...

193 13:44:53.775740 192.168.68.65 192.168.68.62 TCP 78 7000 → 51967 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=64 TSval=10...

Internet Protocol Version 4, Src: 192.168.68.65, Dst: 192.168.68.62

Transmission Control Protocol, Src Port: 7000, Dst Port: 51967, Seq: 0, Ack: 1, Len: 0

Source Port: 7000

Destination Port: 51967

[Stream index: 10]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 3351256260

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 18317346

1011 = Header Length: 44 bytes (11)

Flags: 0x012 (SYN, ACK)

000. = Reserved: Not set

...0 = Accurate ECN: Not set

....0... = Congestion Window Reduced: Not set

....0. = ECN-Echo: Not set

....00. = Urgent: Not set

....01 = Acknowledgment: Set

....0... = Push: Not set

....00. = Reset: Not set

....0...1. = Syn: Set

....0...0. = Fin: Not set

[TCP Flags:A..S.]

Window: 65535

[Calculated window size: 65535]

Checksum: 0xd108 [unverified]

[Checksum Status: Unverified]

Internet Protocol Version 4

Version 4

Header 20

Differentiated Service... 0x00

Total Length 64

Identification 0x0000 (0)

Flags 0x2

Fragment Offset 0

Time to Live 64

Protocol TCP

Header Checksum 0x30e8

Source Address 192.168.68.65

Destination Address 192.168.68.62

6. What is the sequence number of the TCP segment containing the HTTP POST command? Note that in order to find the POST command, you'll need to dig into the packet content field at the bottom of the Wireshark window, looking for a segment with a "POST" within its DATA field.

The sequence number is 1 of the TCP segment containing the HTTP POST command. This sequence number is a relative resequence number found in the TCP segment with the PSH, and ACK flags which contain the HTTP POST request, you can see it from the very bottom.

Time Source Destination Protocol Length Info

44 15:01:45.185029 192.168.68.65 142.250.69.206 TLSv1.2 147 Application Data

45 15:01:45.185118 192.168.68.65 142.250.69.206 TLSv1.2 105 Application Data

46 15:01:45.185144 192.168.68.65 142.250.69.206 TLSv1.2 529 Application Data

47 15:01:45.185553 192.168.68.65 128.119.245.12 TCP 779 61821 → 80 (PSH, ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

48 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

49 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

50 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

51 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

52 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

53 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

54 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

55 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

56 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

57 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

58 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

59 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

60 15:01:45.185553 128.119.245.12 192.168.68.65 TCP 779 80 → 61821 (ACK) Seq=1 Ack=1 Win=2058 Len=713 TSval=1811430426 TSecr=...

Urgent Point

Options: (12)

TCP Option

Kind: No

TCP Option

Kind: No

TCP Option

Kind: T

Length: 0010

Timestamp

0020 f5 0c f1 7d 00 50 1a 11 50 86 5a de cc 68 80 18

Timestamp

0030 00 0a 06 0b 00 00 01 01 08 0a 6b f0 3c 1a 50 53

[Timestamps]

0040 85 c4 50 4f 53 54 20 2f 77 69 72 65 73 68 61 72

[Time since

0050 8b 2d 6c 61 62 73 2f 6c 61 62 33 2d 31 2d 72 65

[Time since

0000 78 8c b5 9f 6a ac 12 56 2d e7 53 95 08 00 45 00

0010 02 fd 00 00 40 00 40 06 bd 8d c0 a8 44 41 80 77

0020 f5 0c f1 7d 00 50 1a 11 50 86 5a de cc 68 80 18

0030 00 0a 06 0b 00 00 01 01 08 0a 6b f0 3c 1a 50 53

0040 85 c4 50 4f 53 54 20 2f 77 69 72 65 73 68 61 72

0050 8b 2d 6c 61 62 73 2f 6c 61 62 33 2d 31 2d 72 65

x...j...V - S...E

...@...DA w

...P...P.Z...h

...k< PS

..POST / wireshar

k-labs/l ab3-1-re

Destination TPLink_9f:6a:ac

Source 12:56:2d:e7:53:95

7. Consider the TCP segment containing the HTTP POST as the first segment in the TCP connection. What are the sequence numbers of the first six segments in the TCP connection (including the segment containing the HTTP POST)? At what time was each segment sent? When was the ACK for each segment received? Given the difference between when each TCP segment was sent, and when its acknowledgement was received, what is the RTT value for each of the six

segments? What is the EstimatedRTT value (see Section 3.5.3, page 242 in text) after the receipt of each ACK? Assume that the value of the EstimatedRTT is equal to the measured RTT for the first segment and then is computed using the EstimatedRTT equation on page 242 for all subsequent Segments.

The sequence numbers of 1-6 segments are: 1, 714, 2162, 3610, 5058, and 6506.

The first 6 segments are #129-134 and the ACK of segments is 140, 141, 146, 147, 148, 149

Segments	Sent time (time of day)	ACK receive time	RTT(seconds)
1	20:44:26.668338	20:44:26.760500	0.092162
2	20:44:26.668505	20:44:26.760502	0.091997
3	20:44:26.668506	20:44:26.763978	0.095472
4	20:44:26.668507	20:44:26.763981	0.095474
5	20:44:26.668507	20:44:26.763983	0.095476
6	20:44:26.668508	20:44:26.763984	0.095476

No.	Time	Source	Destination	Protocol	Length	Info
287	0.000013	192.168.68.64	128.119.245.12	HTTP	1092	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
316	0.000002	128.119.245.12	192.168.68.64	HTTP	843	HTTP/1.1 200 OK (text/html)

Transmission Control Protocol, Src Port: 61954, Dst Port: 80, Seq: 148410, Ack: 1, Len: 1026				Ethernet			
[104 Reassembled TCP Segments (149435 bytes): #129(713), #130(1448), #131(1448), #132(1448), #133(1448), #134(1448), #135(1448)]				Destination TPLink_9f:6a:ac			
[Frame: 129, payload: 0-712 (713 bytes)]				Source 12:56:2d:e7:53:95			
[Frame: 130, payload: 713-2160 (1448 bytes)]				Type IPv4			
[Frame: 131, payload: 2161-3608 (1448 bytes)]				Internet Protocol Version 4			
[Frame: 132, payload: 3609-5056 (1448 bytes)]				Version 4			
[Frame: 133, payload: 5057-6504 (1448 bytes)]				Header ... 20			
[Frame: 134, payload: 6505-7952 (1448 bytes)]				Differentiated Service...			
[Frame: 135, payload: 7953-9400 (1448 bytes)]				Identification 0x0000 (0)			
[Frame: 136, payload: 9401-10848 (1448 bytes)]				Flags 0x2			
[Frame: 137, payload: 10849-12296 (1448 bytes)]				Time to Live 64			
[Frame: 138, payload: 12297-13744 (1448 bytes)]				Protocol TCP			
[Frame: 143, payload: 13745-15192 (1448 bytes)]							
[Frame: 144, payload: 15193-16640 (1448 bytes)]							
[Frame: 145, payload: 16641-18088 (1448 bytes)]							
[Frame: 153, payload: 18089-19536 (1448 bytes)]							
[Frame: 154, payload: 19537-20984 (1448 bytes)]							
[Frame: 155, payload: 20985-22432 (1448 bytes)]							
[Frame: 156, payload: 22433-23880 (1448 bytes)]							
[Frame: 157, payload: 23881-25328 (1448 bytes)]							
[Frame: 158, payload: 25329-26776 (1448 bytes)]							
[Frame: 159, payload: 26777-28224 (1448 bytes)]							
[Frame: 160, payload: 28225-29672 (1448 bytes)]							
[Frame: 161, payload: 29673-31120 (1448 bytes)]							

129	20:44:26.668338	192.168.68.64	128.119.245.12	TCP	779	61954 → 80 [PSH, ACK] Seq=1 Ack=1 Win=2058 Len=713
130	20:44:26.668505	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80 [ACK] Seq=714 Ack=1 Win=2058 Len=1448 TS
131	20:44:26.668506	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80 [ACK] Seq=2162 Ack=1 Win=2058 Len=1448 T
132	20:44:26.668507	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80 [ACK] Seq=3610 Ack=1 Win=2058 Len=1448 T
133	20:44:26.668507	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80 [ACK] Seq=5058 Ack=1 Win=2058 Len=1448 T
134	20:44:26.668508	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80 [ACK] Seq=6506 Ack=1 Win=2058 Len=1448 T
135	20:44:26.668509	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80 [ACK] Seq=7954 Ack=1 Win=2058 Len=1448 T
136	20:44:26.668510	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80 [ACK] Seq=9402 Ack=1 Win=2058 Len=1448 T
137	20:44:26.668511	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80 [ACK] Seq=10850 Ack=1 Win=2058 Len=1448

ACK of segments:

139	20:44:26.760497	128.119.245.12	192.168.68.64	TCP	74	80 → 61959	[SYN, ACK] Seq=0 Ack=1 Win=28960 Len=0 MSS=14
140	20:44:26.760500	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=714 Win=238 Len=0 TSval=15410
141	20:44:26.760502	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=2162 Win=261 Len=0 TSval=1541
142	20:44:26.760724	192.168.68.64	128.119.245.12	TCP	66	61959 → 80	[ACK] Seq=1 Ack=1 Win=131712 Len=0 TSval=5742
143	20:44:26.760823	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK] Seq=13746 Ack=1 Win=2058 Len=1448 TSval
144	20:44:26.760864	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK] Seq=15194 Ack=1 Win=2058 Len=1448 TSval
145	20:44:26.760867	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK] Seq=16642 Ack=1 Win=2058 Len=1448 TSval
146	20:44:26.763978	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=3610 Win=283 Len=0 TSval=1541
147	20:44:26.763981	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=5058 Win=306 Len=0 TSval=1541
148	20:44:26.763983	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=7954 Win=351 Len=0 TSval=1541
149	20:44:26.763984	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=9402 Win=374 Len=0 TSval=1541
150	20:44:26.763986	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=10850 Win=396 Len=0 TSval=154
151	20:44:26.763987	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=12298 Win=419 Len=0 TSval=154
152	20:44:26.763988	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK] Seq=1 Ack=13746 Win=442 Len=0 TSval=154
153	20:44:26.764163	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK] Seq=18090 Ack=1 Win=2058 Len=1448 TSval
154	20:44:26.764169	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK] Seq=19538 Ack=1 Win=2058 Len=1448 TSval

The formula given: $\text{EstimatedRTT} = 0.875 * \text{EstimatedRTT} + 0.125 * \text{SampleRTT}$

EstimatedRTT after the receipt of the ACK:

Segment 1: EstimatedRTT = RTT for segment 1 = 0.092162 s

Segment 2: EstimatedRTT = $0.875 * 0.092162 + 0.125 * 0.091997 = 0.090482459$

Segment 3: EstimatedRTT = $0.875 * 0.090482459 + 0.125 * 0.095472 = 0.09110615162$

Segment 4: EstimatedRTT = $0.875 * 0.09110615162 + 0.125 * 0.095474 = 0.09165213267$

Segment 5: EstimatedRTT = $0.875 * 0.09165213267 + 0.125 * 0.095476 = 0.09213011608$

Segment 6: EstimatedRTT = $0.875 * 0.09213011608 + 0.125 * 0.095476 = 0.09254835157$

8. What is the length of each of the first six TCP segments?

The first segment is 713 bytes, the other 5 are the same and have a value of 1448 bytes

No.	Time	Source	Destination	Protocol	Length	Info
287	0.000013	192.168.68.64	128.119.245.12	HTTP	1092	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
316	0.000002	128.119.245.12	192.168.68.64	HTTP	843	HTTP/1.1 200 OK (text/html)

Transmission Control Protocol, Src Port: 61954, Dst Port: 80, Seq: 148410, Ack: 1, Len: 1026		Ethernet	
[104 Reassembled TCP Segments (149435 bytes): #129(713), #130(1448), #131(1448), #132(1448), #133(1448), #134(1448), #135(1448)]		[0] [15] [16]	
[Frame: 129, payload: 0-712 (713 bytes)]		Destination TPLink_9f:6a:ac	
[Frame: 130, payload: 713-2160 (1448 bytes)]		Source 12:56:2d:e7:53:95	
[Frame: 131, payload: 2161-3608 (1448 bytes)]		Type IPv4	
[Frame: 132, payload: 3609-5056 (1448 bytes)]		Internet Protocol Version 4	
[Frame: 133, payload: 5057-6504 (1448 bytes)]		[0] [15] [16]	
[Frame: 134, payload: 6505-7952 (1448 bytes)]		Version 4	
[Frame: 135, payload: 7953-9400 (1448 bytes)]		Header ... 20	
[Frame: 136, payload: 9401-10848 (1448 bytes)]		Differentiated Service... 0x00	
[Frame: 137, payload: 10849-12296 (1448 bytes)]		Identification 0x0000 (0)	
[Frame: 138, payload: 12297-13744 (1448 bytes)]		Flags 0x2	
[Frame: 139, payload: 13745-15192 (1448 bytes)]		Time to Live 64	
[Frame: 140, payload: 15193-16640 (1448 bytes)]		Protocol TCP	
[Frame: 141, payload: 16641-18088 (1448 bytes)]			
[Frame: 142, payload: 18089-19536 (1448 bytes)]			
[Frame: 143, payload: 19537-20984 (1448 bytes)]			
[Frame: 144, payload: 20985-22432 (1448 bytes)]			
[Frame: 145, payload: 22433-23880 (1448 bytes)]			
[Frame: 146, payload: 23881-25328 (1448 bytes)]			
[Frame: 147, payload: 25329-26776 (1448 bytes)]			
[Frame: 148, payload: 26777-28224 (1448 bytes)]			
[Frame: 149, payload: 28225-29672 (1448 bytes)]			
[Frame: 150, payload: 29673-31120 (1448 bytes)]			

9. What is the minimum amount of available buffer space advertised at the received for the entire trace? Does the lack of receiver buffer space ever throttle the Sender?

The minimum amount of available buffer advertised at the received for the entries trace is 28960 bytes; According to the trace, the sender is never throttled due to lacking receiver buffer space.

128	20:44:26.662822	192.168.68.64	128.119.245.12	TCP	78	61959 → 80	[SYN]	Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=574271163 TSecr=
129	20:44:26.668338	192.168.68.64	128.119.245.12	TCP	779	61954 → 80	[PSH, ACK]	Seq=1 Ack=1 Win=2058 Len=713 TSval=2757783341 TSecr=
130	20:44:26.668505	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=714 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
131	20:44:26.668506	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=2162 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
132	20:44:26.668507	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=3610 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
133	20:44:26.668507	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=5058 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
134	20:44:26.668508	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=6506 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
135	20:44:26.668509	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=7954 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
136	20:44:26.668510	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=9402 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
137	20:44:26.668511	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=10850 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
138	20:44:26.668512	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=12298 Ack=1 Win=2058 Len=1448 TSval=2757783341 TSecr=
139	20:44:26.760497	128.119.245.12	192.168.68.64	TCP	74	80 → 61959	[SYN, ACK]	Seq=0 Ack=1 Win=28960 Len=0 MSS=1460 SACK_PERM TSv
140	20:44:26.760500	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK]	Seq=1 Ack=714 Win=238 Len=0 TSval=1541025097 TSecr=2757
141	20:44:26.760502	128.119.245.12	192.168.68.64	TCP	66	80 → 61954	[ACK]	Seq=1 Ack=2162 Win=261 Len=0 TSval=1541025097 TSecr=275
142	20:44:26.760724	192.168.68.64	128.119.245.12	TCP	66	61959 → 80	[ACK]	Seq=1 Ack=1 Win=131712 Len=0 TSval=574271261 TSecr=1541
143	20:44:26.760823	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=13746 Ack=1 Win=2058 Len=1448 TSval=2757783433 TSecr=
144	20:44:26.760864	192.168.68.64	128.119.245.12	TCP	1514	61954 → 80	[ACK]	Seq=15194 Ack=1 Win=2058 Len=1448 TSval=2757783433 TSecr=

[Stream index: 6]
> [Conversation completeness: Incomplete, ESTABLISHED (7)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 157592656
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 423683146
1010 = Header Length: 40 bytes (10)
> **Flags: 0x012 (SYN, ACK)**
Window: 28960
[Calculated window size: 28960]
Checksum: 0x5cb7 [unverified]
[Checksum shift: 16 bits]

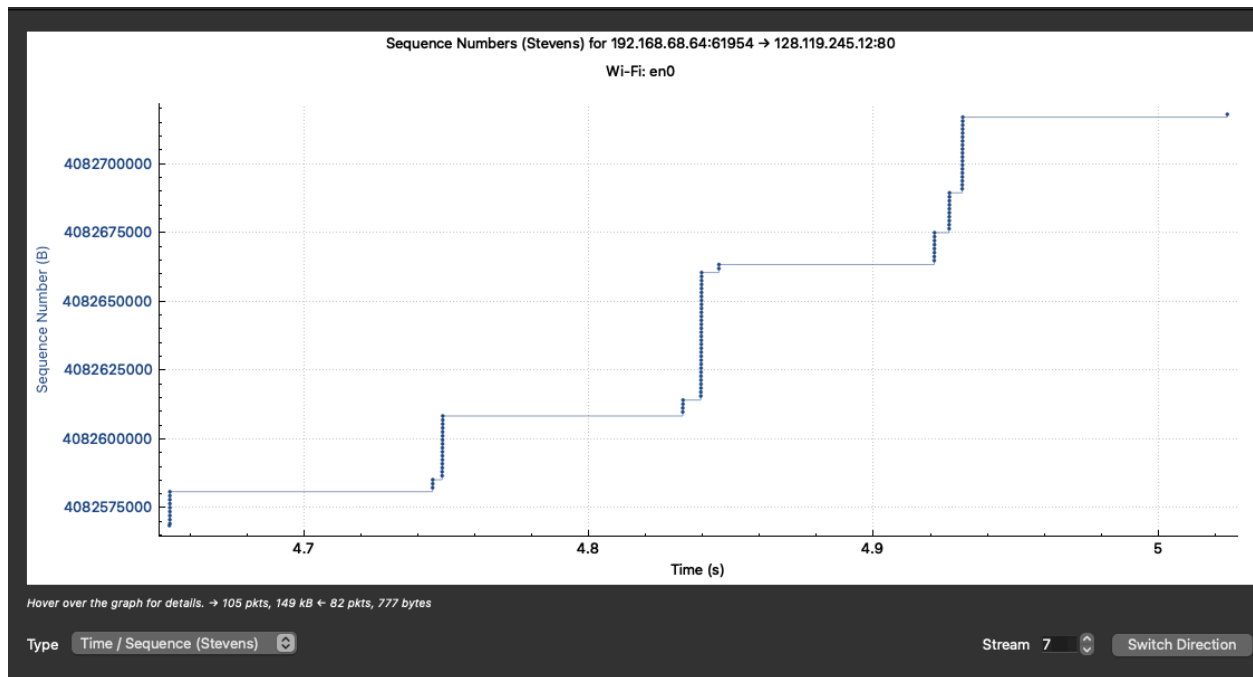
Ethernet
0 15
Destin
12:56:2d:a
Sour
TPLink_8
Type
IPv4

10. Are there any retransmitted segments in the trace file? What did you check for (in the trace) in order to answer this question?

There are no retransmitted segments in the trace file because, in the time sequence graph (Stevens) shown below the sequence numbers are increasing steadily over time, meaning new data is being sent without interruption. No sudden drops or duplicate sequence numbers, which are common indicators of retransmitted segments.

I also did some additional checks firstly, I checked whether a segment with the same sequence number appeared more than once. If it did and was not a normal retransmission due to window size, it would be considered a retransmitted segment. Then, I examined the TCP retransmission flag in Wireshark, where retransmissions are explicitly marked as TCP Retransmission or Fast Retransmission in the "Info" column. Finally, I checked for any fast retransmissions, which occur when the receiver sends three duplicate ACKs (same ACK number repeated). In such cases,

the sender may perform a fast retransmission.

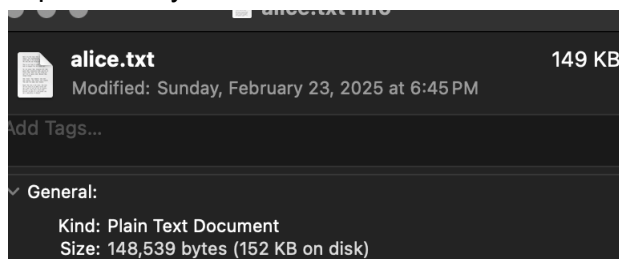


3. Analyze the ACK Pattern (Gaps in Acknowledgments)

11. How much data does the receiver typically acknowledge in an ACK? Can you identify cases where the receiver is ACKing every other received segment (see Table 3.2 on page 250 in the text).

According to the given table in the text book, The receiver usually acknowledges 1460 bytes in an ack. When the segment is ACKing every other the data is doubled.

12. What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value.



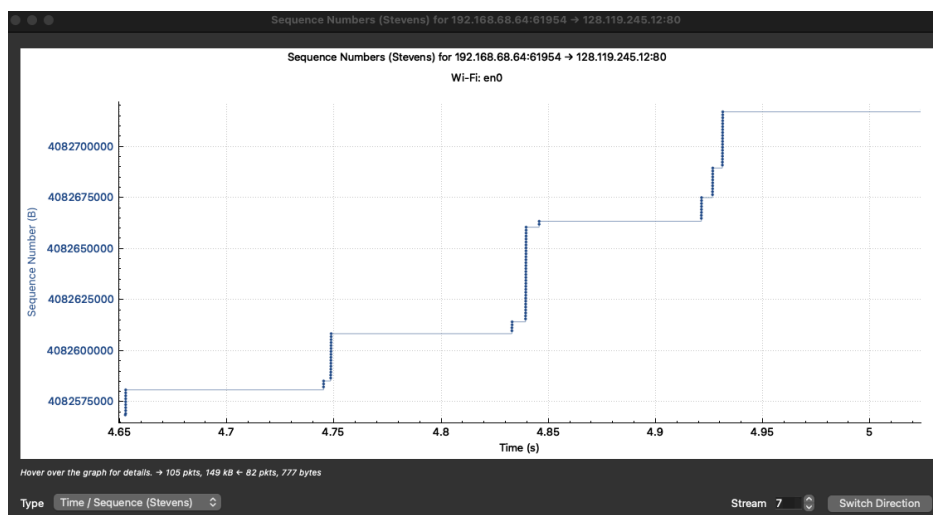
The alice.txt on my laptop is 148,539 bytes,

Download time = 1.574736000 (First TCP segment) - 0.261257000 seconds (last ACK) = 1.313479 seconds.

Throughput = 148,539 bytes/1.313479seconds = 113,054.84 bytes/second

4. TCP congestion control in action

13. Use the Time-Sequence-Graph(Stevens) plotting tool to view the sequence number versus time plot of segments being sent from the client to the gaia.cs.umass.edu server. Can you identify where TCP's slowstart phase begins and ends, and where congestion avoidance takes over? Comment on ways in which the measured data differs from the idealized behavior of TCP that we've studied in the text.



The slow start grows exponentially, from the graph above we really cannot tell where the slow start is, however, if the slow start has happened, we should see an exponentially increasing pattern. After a slow start, the sequence number increase becomes more linear, indicating that TCP has transitioned to congestion avoidance mode. In this mode, TCP increases the window size additively rather than exponentially, meaning the data transfer rate grows more gradually. In the graph, we can see that the sequence number continues to increase but at a steady rate, rather than the sharp upward jumps seen in slow start. Thus, I can't tell where congestion avoidance takes over.

14. Answer each of the two questions above for the trace that you have gathered when you transferred a file from your computer to gaia.cs.umass.edu

Each of the two questions is answered